TrustCharge: AI-Based Centralized EV Charging Reliability and Smart Reservation Platform

Computer Science and Engineering

ALEESHA M A
CHN23CS030

JIA CHINNU ABRAHAM
CHN23CS101

JOEL GEORGE MATHEW
CHN23CS102

LOIS ROSE DAVID
CHN23CS120

Guide: Prof. Gopakumar G

Department of Computer Engineering
College of Engineering Chengannur

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Overview

- Unifies fragmented Electric Vehicle (EV) charging networks into a single, seamless ecosystem.
- Integrates live NDW, TomTom and OpenChargeMap data .
- Verifies EVs through RDW and reads live battery and range via Smartcar telemetry.
- Reserves charging slots with wallet payments ,Plus and Pro memberships.
- Recommends stations with a Gemini-powered assistant using battery, range and distance.

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Problem Statement

Problem

To address the unreliability of existing EV charging systems that provide inaccurate availability information, causing long waiting times, failed charging attempts, and severe range anxiety.

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Literature Review

No.	Paper (Year)	Methodology	Gap identified
1	Multi-Objective EV Charging Station Allocation (2023)	Multi-objective optimization of station and renewable placement	No real-time booking, trust evaluation or predictive availability
2	Integration of Electric Vehicle into Smart Grid (2024)	Improved Honey Badger Algorithm for V2G and G2V	No slot reservation, real-time availability or prepaid booking
3	Realizing Smart Charging of EVs at Public Charging Infrastructures (2024)	ISO 15118 communication architecture at a charging hub	No reservation, route planning or reliability prediction
4	AI-Based EV Smart Charging System in Malaysia (2024)	ML, predictive analytics, demand response with V2G	Limited deployment and local datasets; no large-scale AI

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Literature Review

No.	Paper (Year)	Methodology	Gap identified
5	Distributing Charging Requests: A Reservation-Based Approach (2023)	Book a station before arrival; tested on real data	Advance booking needed; no AI waiting-time or demand prediction
6	Reserving Charging Decision-Making Model and Route Plan (2018)	K-Shortest Path selection by traffic, distance, cost, availability	No AI/ML; only route planning and reservation
7	EV Charging: A Business Intelligence Model (2025)	AI/ML on EV and station data to recommend schedules	Needs large real-time data; hard to scale; secure sharing required
8	Long Trip Charging Planning with Waiting Time Forecast (2025)	MIDP with Monte Carlo simulation	Fixed traffic assumptions; no real-time traffic or dynamic pricing

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Identified Gaps

- No single system combines AI-based reliability prediction with cross-network reservation.
- Existing reservation systems do not use AI to forecast waiting time or demand.
- AI and optimization approaches rarely consider real-time slot booking or prepaid access.
- Centralized multi-provider platforms remain vulnerable, with limited scalability.
- Real-world, large-scale validation and real-time traffic integration are largely missing.
- No unified membership or wallet system spans multiple charging networks.



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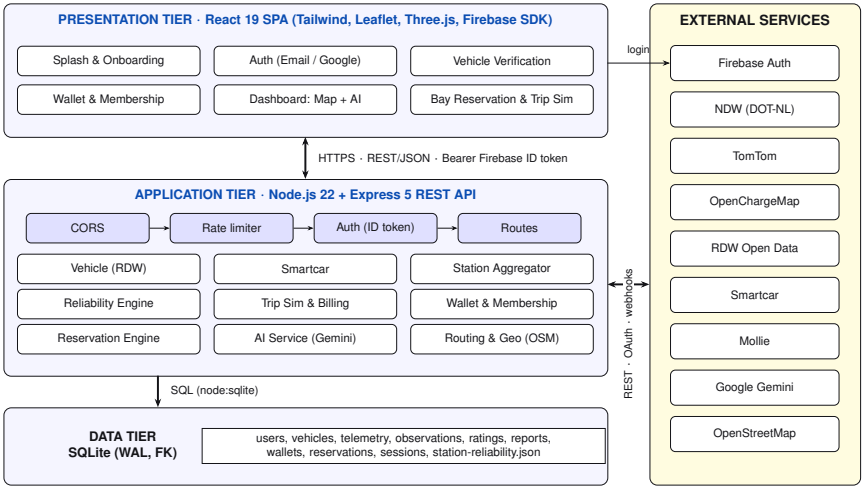
Proposed Solution

Objectives

- Unify fragmented EV charging networks in one centralized application.
- Recommend stations with an AI assistant using live battery, range and distance.
- Enable wallet-paid bay reservation with plan limits and cancellation windows.
- Offer one wallet and membership system (Plus, Pro) with Mollie payments.
- Provide membership benefits: fee and energy discounts, more bookings, longer book-ahead.
- Reduce range anxiety through telemetry, OSRM routing and data-driven picks.
- Scale to onboard new charging providers and regions.

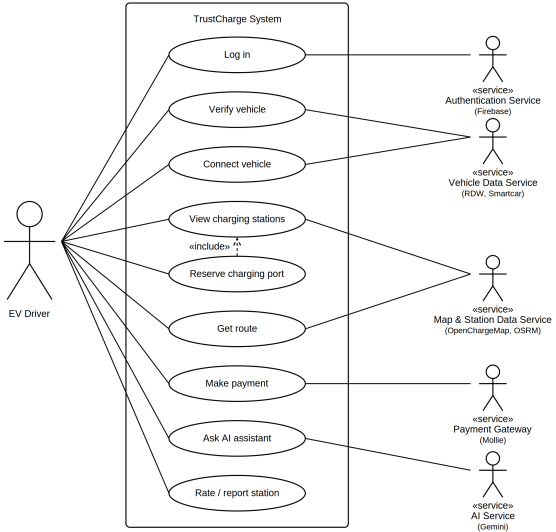
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System Architecture



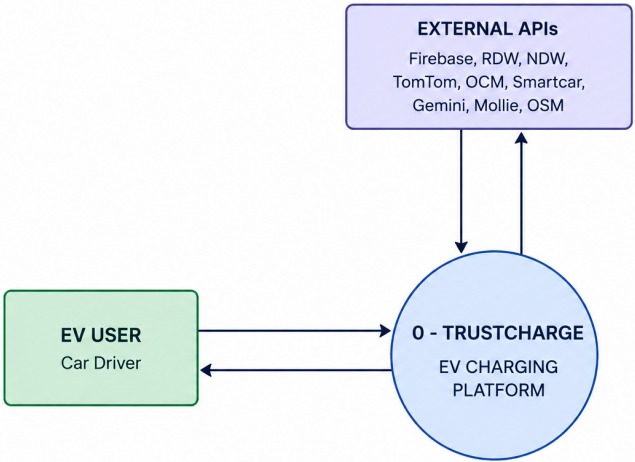
Use Case Diagram

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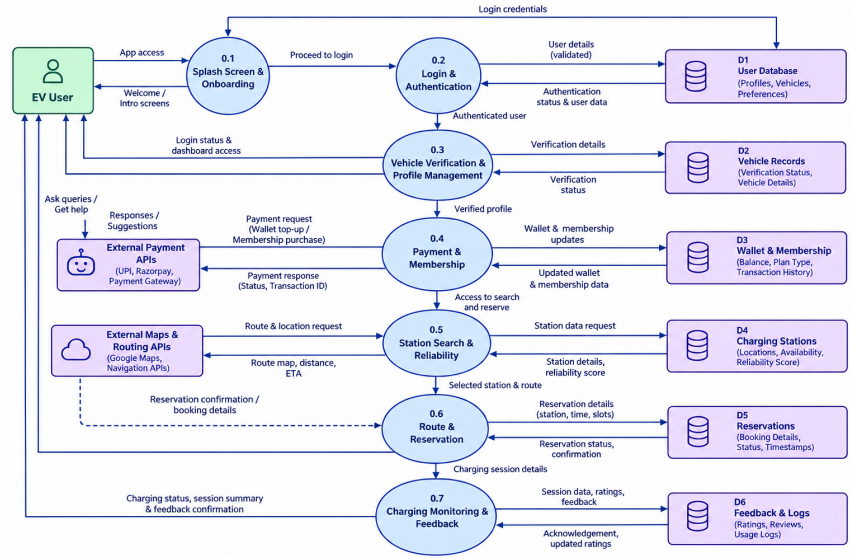
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Data Flow Diagram – Level 0



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Data Flow Diagram – Level 1



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Methodology

1. **Authenticate and verify:** Firebase sign-in; RDW plate check confirms an EV; Smartcar gives battery, range, position.
2. **Collect live data:** It collect live station data from NDW, TomTom and OpenChargeMap, refreshed every 5 minutes.
3. **Smart Recommendation:** The system recommends a charging station, choosing exclusively from available stations that fall within the car's current live battery range.
4. **Route Planning:** OSRM calculates the route, and the planner selects the best reachable station at each step while maintaining a 10% battery reserve.
5. **Reserve, pay, and bill:** The driver books a slot from the wallet under plan limits. Mollie handles memberships and top-ups, and the trip is billed with the member discount.

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Benefit	Plus	Pro
Price	€4.99	€9.99
Wallet credit on purchase	€10	€25
Reservation fee discount	50%	100%
Energy discount	5%	12%
Active reservations	2	4
Book ahead	72 h	7 days
Max duration	90 min	120 min
Free-cancel window	30 min	10 min

- Fee is €1 per 30 minutes, reduced by the plan discount and billed from the wallet; timely cancellation refunds it.
- Plus and Pro memberships are purchased through Mollie and remain active for 30 days.
- Nearby stations are ranked using a distance-based penalty; Gemini selects from the provided list.

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Functional Requirements

- Register and log in with Firebase Authentication (email or Google).
- Verify an EV by number plate using RDW data and save the vehicle.
- Link a car through Smartcar; read battery, range and position.
- Show live stations on a map with availability, tariffs, connector and power.
- Let users rate stations and report faults.
- Answer questions and recommend stations with a Gemini assistant.
- Plan routes, geocode places and list nearby restaurants.
- Sell Plus / Pro plans and wallet top-ups through Mollie with webhook confirmation.
- Reserve a slot within plan limits.
- Simulate a trip with charging stops and bill energy from the wallet.

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Non-Functional Requirements

Quality	How it is met
Security & abuse protection	Firebase ID-token checks on the API, Smartcar tokens encrypted at rest, signed OAuth state, per-minute rate limits
Data integrity	Idempotent Mollie crediting, server-side cost recalculation, wallet never below €0, foreign keys and WAL
Performance	Short-lived caches: 15 s station snapshots, 2 min TomTom availability, 10 min tariffs
Fault tolerance	Gemini fallback model and local scoring, Overpass mirrors, TomTom fallback for restaurants
Usability	Map-first dashboard, guided onboarding, in-app AI assistant and trip companion
Scalability & transparency	Provider-agnostic station layer for new networks; confidence shown, no fabricated reliability

Hardware Requirements

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Server (minimum)

Component	Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB minimum
Disk	50 GB available space
Network	Internet connection for APIs

Client Devices

- Laptop, desktop or phone with a modern web browser.
- Location access for nearby station search.
- Smartcar-compatible EV for live telemetry; simulated vehicles are used for testing.

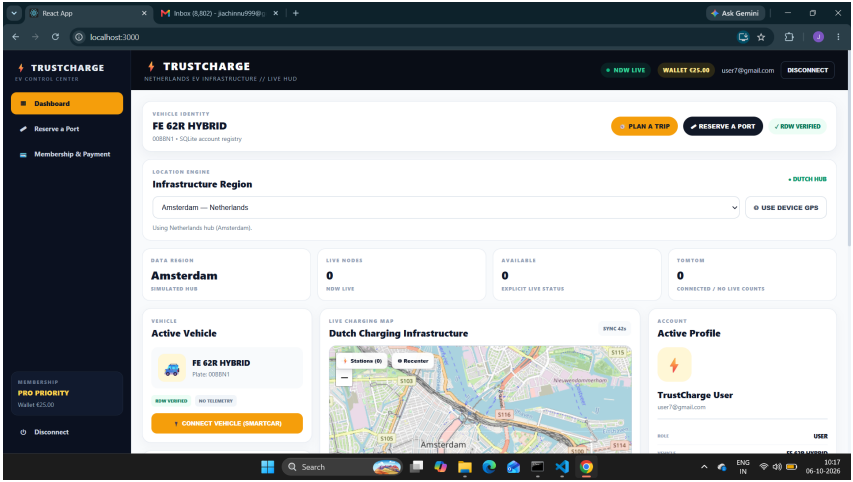
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Software Requirements

Layer	Technologies
Frontend	React 19, Tailwind CSS 4, Leaflet / react-leaflet, Three.js, Lucide icons, Axios, Firebase SDK
Backend	Node.js 22+ (built-in node:sqlite), Express 5, SQLite, dotenv, CORS, @google/genai, firebase-admin (optional)
Services & APIs	Firebase Authentication, NDW, TomTom, OpenChargeMap, RDW, Smartcar, Mollie, Google Gemini, OpenStreetMap (Nominatim, OSRM, Overpass)
Tools & platform	VS Code, Git and GitHub, npm; Chrome, Firefox, Edge or Safari; Windows, Linux or macOS

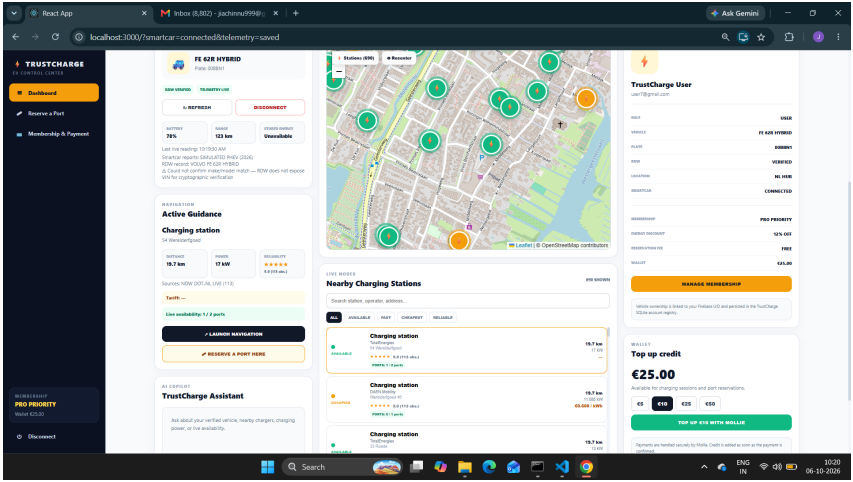
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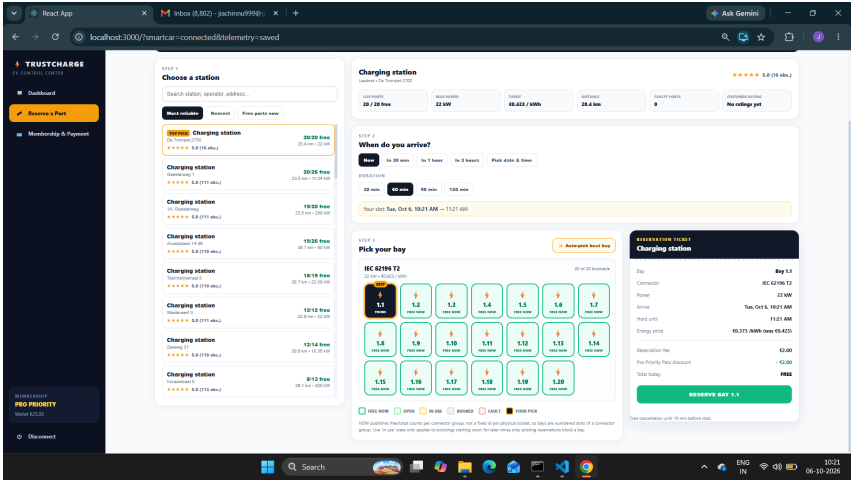
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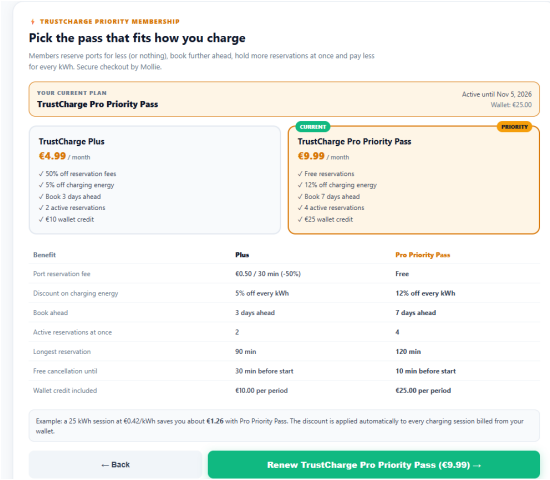
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React App
Inbox (8,802) - jiachinru999@
Ask Gemini
localhost:3000/?smartcar=connected&telemetry=saved
TRUSTCHARGE
EV CONTROL CENTER
Dashboard
Reserve a Port
Membership & Payment
Disconnect
STEP 1
Choose a station
Search station, operator, address...
Most reliable
Nearest
Free parts now
Charging station
Charging station
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TRUSTCHARGE PRIORITY MEMBERSHIP

Pick the pass that fits how you charge

Members reserve ports for less (or nothing), book further ahead, hold more reservations at once and pay less for every kWh. Secure checkout by Mollie.

YOUR CURRENT PLAN
TrustCharge Pro Priority Pass
Active until Nov 5, 2026
Wallet: €25.00

TrustCharge Plus
€4.99 / month
- ✓ 50% off reservation fees
- ✓ 5% off charging energy
- ✓ Book 3 days ahead
- ✓ 2 active reservations
- ✓ €10 wallet credit

CURRENT · PRIORITY
TrustCharge Pro Priority Pass
€9.99 / month
- ✓ Free reservations
- ✓ 12% off charging energy
- ✓ Book 7 days ahead
- ✓ 4 active reservations
- ✓ €25 wallet credit

Benefit	Plus	Pro Priority Pass
Port reservation fee	€0.50 / 30 min (-50%)	Free
Discount on charging energy	5% off every kWh	12% off every kWh
Book ahead	3 days ahead	7 days ahead
Active reservations at once	2	4
Longest reservation	90 min	120 min
Free cancellation until	30 min before start	10 min before start
Wallet credit included	€10.00 per period	€25.00 per period

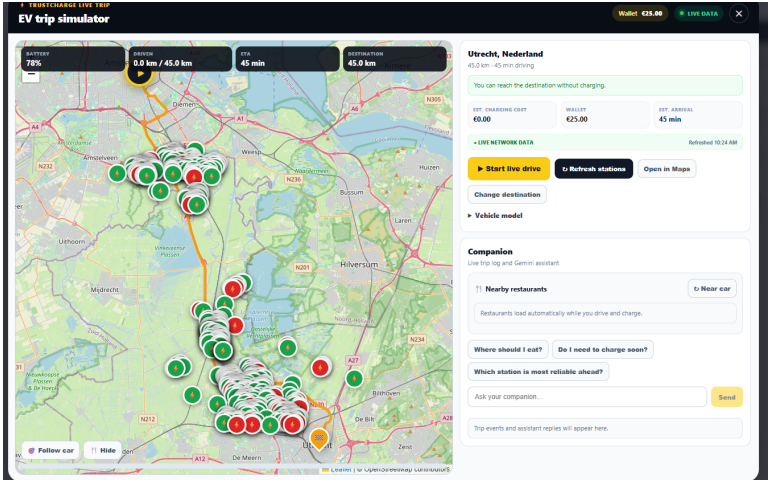
Example: a 25 kWh session at €0.42/kWh saves you about **€1.26** with Pro Priority Pass. The discount is applied automatically to every charging session billed from your wallet.

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Renew TrustCharge Pro Priority Pass (€9.99) →

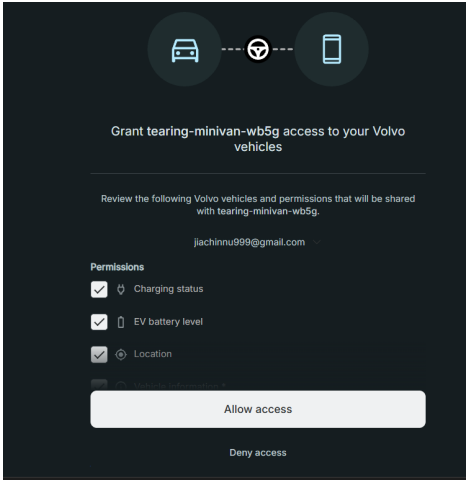
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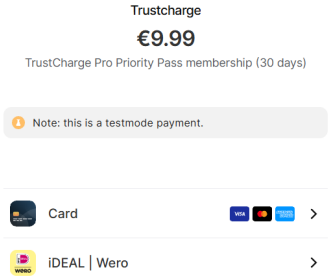
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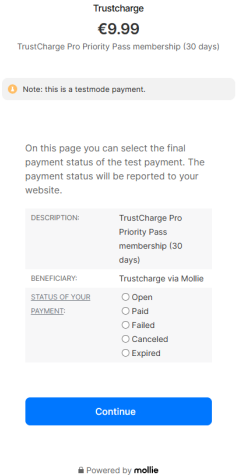
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Conclusion

- TrustCharge is a centralized EV charging platform that unifies live station data, reservations, wallet payments and vehicle telemetry.
- **One platform:** live data from NDW, TomTom and OpenChargeMap in a single map.
- **One wallet:** Mollie-backed memberships, top-ups and bay reservations across networks.
- **Honest AI:** Gemini picks only from real stations; reliability is never fabricated.
- AI-based reliability and queue-time prediction remain future work, building on the data now being collected.



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- **AI charger reliability prediction:** train a model on station observations to forecast whether a charger works on arrival.
- **AI queue-time prediction** using port occupancy, arrival patterns and logged queue minutes.
- **Longer data collection:** 3,990 samples across 1,574 stations is too thin to train on.
- Feed predicted reliability and queue time into ranking and Gemini recommendations.
- Move from simulated to live Smartcar vehicles and validate with real users at scale.
- Onboard more providers and regions beyond the Netherlands.

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- Base Paper (Long Trip Charging Planning)
- TrustCharge: Reliability-Aware Long-Trip Charging Planning and Port Reservation
- AI-Based Smart Electric Vehicle Charging: A Review of Load Balancing, Reservation, Reliability and Waiting-Time Prediction
- Funding Proposal



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